Structure and Crack Response to Coal Blasting in Brazil

Rosenhaim, V.L., Aimone-Martin, C.T. Munaretti, E., Koppe, J.C.

Abstract

Blasting near residential areas has become frequent in many locations throughout Brazil. It was deemed necessary by the authors to have a better understanding of how typical Brazilian structures respond to blast vibrations and compare this response to environmental forces that naturally act on these structures. Typical Brazilian structures are built with clay bricks and cement-mortar. Foundations comprise concrete beam perimeter walls and thin concrete slabs. Roofs are very heavy as they are construction with clay tiles over a wood frame. Typically, walls are cover with cement grout, spackling (or plastered) and then painted. Interior walls are made of clay bricks and mortar to divide in rooms inside the structures.

Velocity transducers were placed within a structure to measure whole structure and mid-wall motions in order to compute induced global wall strains. Displacement gages were mounted over an existing wall crack and on uncracked wall material to measure changes in crack width resulting from daily variations in weather conditions and blast-induced ground motions. A clay brick and mortar structure located inside the operation area of a coal mine was subjected to daily blasts at distances ranging from 32 to 810 m (105 to 2657 ft) and charge weights per delay varying from approximately 14 to 250 kg (30 to 550 lbs). Scaled distances ranged from 3.5 to 122.6 m/kg$^{1/2}$ (7.7 to 268.4 ft/lb$^{1/2}$). A total of 115 blasts were recorded during the six month period of the study.

Results of long-term, daily variations in temperature and humidity affect crack displacements greater than the influence of blasting-induced ground motions. Weather-induced overall crack aperture opening and closing was approximately 2.7 times greater than the maximum blast-induced crack displacements. Computed maximum in-plane wall strains were 1.1 times smaller than the failure strains for the weakest material comprising the structure walls, while maximum mid-wall bending strain was 2 times smaller than the failure strains reported in literature. This indicates that weather affects are more likely the cause of the structures material fatigue leading to cosmetic damage than blasting.

Introduction

The effects of ground vibrations and air overpressure (airblast) are often perceived by communities neighboring mining operations, quarries and construction areas where explosives are employed to fragment rock. With today's urban expansion, many mining operations that were isolated in the past are now surrounded by residential communities and this situation is common all over the world. There are also cases where blasting is required inside urban areas near buildings and residences. Activities related to mining and blasting have become a problem as they generate a potential nuisance to nearby communities resulting from high air overpressures and can result in claims of wall cracks associates with ground vibrations.

In Brazil, there have been no studies measuring the influence of ground vibrations and air overpressures in structures typical of national construction. Aiming to fill this gap, a series of structure response studies were initiated to obtain a better understanding of typical Brazilian structure's response to blasting. This study presents the results of the first investigation conducted in a residential structure located within the operational area of a coal mine in Rio Grande do Sul. The proximity of the structure, owned by the mining company, to the operation allowed blasting to take place very close-in, approximately 30 m (100 ft) from the house.

Blasting in open pit coal mines in Brazil is conducted using low powder factors, ranging from 150 g/m^3 (0.253 lb/yd^3) in overburden blasts, to 270 g/m^3 (0.455 lb/yd^3) in coal seams shots, resulting in very confined blasts with little horizontal and vertical movement. These confined blasts can lead to high levels of ground vibrations. Blasting is required in different bench levels where mechanical excavation is not possible. The average blast hole length is 5 m (16.4 ft) drilled in a 76 mm (3 in) diameter arranged in staggered patterns.

Figure 1 shows the location of the velocity transducers in the structure and a location map of the house in relation to the blasting area. Velocity transducers were mounted in the southeast corner of the structure, facing the blasting. Single-axis geophones were installed in the interior of the structure in the upper corner of the walls near the ceiling, in the lower corner, near the foundation, and on the mid-walls facing east and south. A tri-axial geophone was buried in the ground in the exterior close to the instrumented structure corner to record ground vibrations that most likely drove the structure motion. A microphone was place in the same location to record air overpressure levels.

The south wall contained the longitudinal (radial) velocity transducer perpendicular to the wall plane and is also called the radial wall. The transverse wall was the east wall and contained the transverse velocity transducer mounted perpendicular to its plane. Differential displacement between upper structure (*S2*) and lower structure (*S1*) in the transverse direction induced in-plane tensile strains in the south wall, while differential displacements in the radial direction generated in-plane tensile strains in the east wall. Mid-wall bending strains were computed using the velocity transducers mounted in the middle of the walls, as shown in Figure 1.

Shear strains (γ_{max}) were estimated by calculating the differential displacement between upper structure and lower structure ($\Delta\delta_{max} = S2 - S1$), divided by the wall height (H) (Dowding, 2000), expressed in micro-strains (10^{-6} mm/mm) using the following equation:

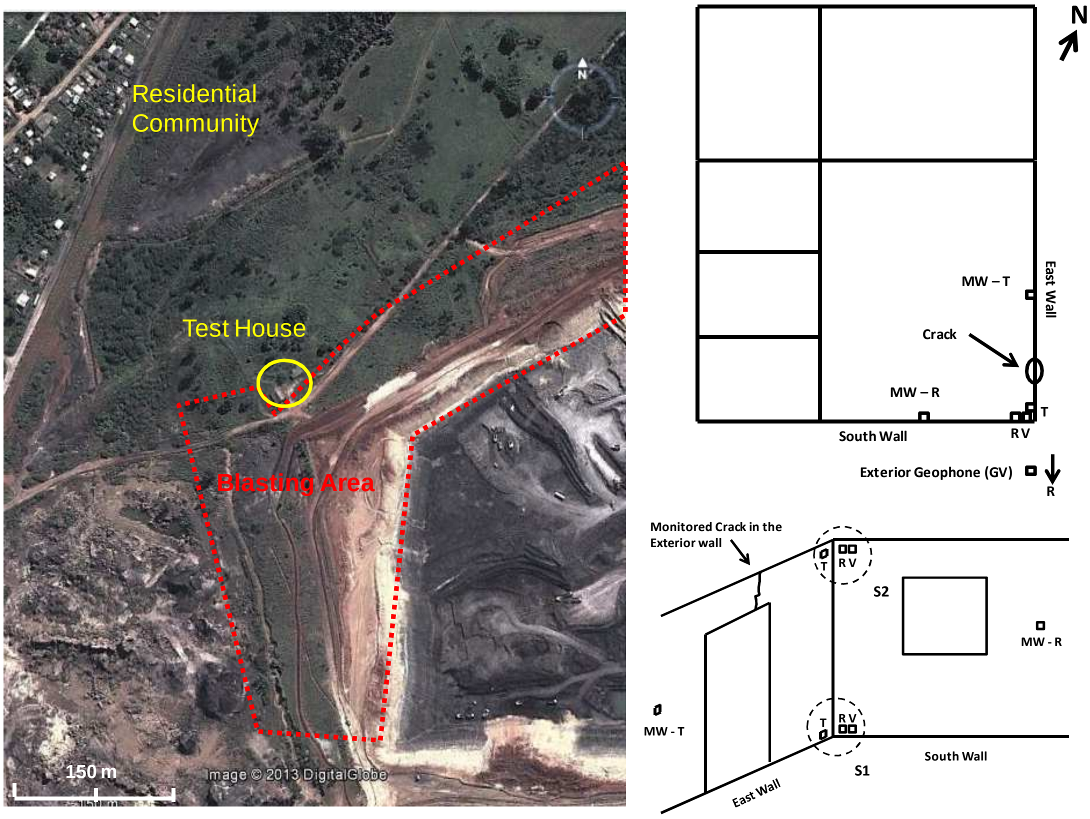


Figure 1: Location map showing the test house and blasting areas (left) and schematic drawing of the structure showing the location of the single-axis velocity transducers and instrumented wall crack (right)

$$\gamma_{max} = \left(\frac{\Delta\delta_{max}}{H}\right)$$

In-plane tensile strains (ε_{Lmax}) is the deformation mode most likely to cause cosmetic cracks in walls during high levels of structure motion and is directly related to global shear strains in the walls through the following equation:

$$\varepsilon_{Lmax} = \gamma_{max}(sin\ \theta)(\cos\theta)$$

Where, θ is the interior angle of the longest diagonal of the wall subjected to deformation relative to the horizontal plane. Theta, θ, is determined by the inverse tangent of the ration of wall height and wall length.

Mid-wall bending strains are directly related to bending stresses induced in the walls. The walls can be modeled as a beam fixed in both ends, in other words, in the foundation (*S1*) and at the roof (*S2*). For structures well couple to the ground *S1* is "fixed". However, the roof can be modeled with various degrees of "coupling", from relatively free to extremely fixed. In the case of the studied structure, the "fixed-fixed" model was considered as the roof is very heavy construction with clay tiles over a wood frame. The model considered is also the one that results in the greatest estimates for wall strains, and is determined by the following equation:

$$\varepsilon = \left(\frac{6d\Delta\delta_{max}}{H^2}\right)$$

Where, ε is the bending strain, d is half of the wall thickness subjected to deformation (in mm). Calculated strains can be compared with the failure limits of each material used in the structure construction, and the probability of causing superficial cracks in the walls can be estimated (Aimone-Martin *et al.*, 2003).

A displacement gage was mounted over an existing exterior wall crack located on the door lintel, shown in Figure 2. This gage was used to measure variations in crack aperture due to blast-induced structure motions. Crack width was correlated with ground vibrations and air overpressure amplitudes and variations resulting from climate changes. Aperture variations were measured on an hourly basis over six months and correlated with daily variations of temperature and humidity (weather-induced crack response). A gage was installed on a portion of undamaged wall (null gage), also shown in Figure 2, to measure the natural expansion and contraction of the material with weather changes and this value was subtracted from the measurements of the crack aperture to isolate crack movement.

Figure 2: Location of displacement gages above a doorway, mounted over the existing wall crack and in the undamaged wall

Results

Over the six months period of study 148 blasts were conducted and 115 events were recorded by the instruments in structure. Charge weights per delay ranged from 13.8 kg (30.4 lb) to 245.9 kg (542.1 lb) averaging 85 kg (187.4 lb). Distances between blasts and the monitored structure ranged from 32 m (105 ft) to 810 m (2657.5 ft), resulting in a range of scaled distances between 3.5 and 122.6 m/kg$^{1/2}$ (7.7 and 268.4 ft/lb$^{1/2}$). The maximum peak particle velocity (PPV) recorded at the structure was 83.31 mm/s (3.28 in/s) at a distance of 33 m (108.3 ft) from the blast, with charge weight per delay of 58.3 kg (128.5 lb). Peak frequencies ranged from 3.9 Hz to 36.5 Hz and averaged 15.5 Hz. From the 115 blasts recorded, eleven presented peak particle velocities above the limits suggested by the Brazilian standard regulation for blasting in urban areas (ABNT NBR 9653:2005) as shown in Figure 3.

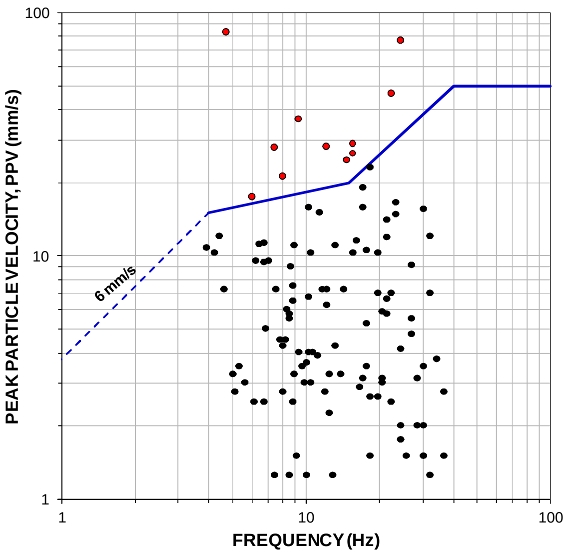


Figure 3: Peak particle velocity versus peak frequency showing data recorded at the test structure plotted in the Brazilian safe blasting standards (ABNT NBR 9653)

Figure 4 presents correlations between peak upper structure (*S2*) and mid-wall (*MW*) horizontal displacements with peak particle velocity and peak ground displacement (designated GV) and air overpressure. Upper structure and mid-wall movement in the direction facing most blasts show higher correlations with peak ground velocities than the perpendicular direction. The correlation coefficients for the *S2-GV* displacement relationships are 97% and 91% with a slope close to one. This indicates the structure is rigid as the upper structure and ground displacements move with similar amplitudes. The mid-wall facing most of the blasting (*MW-R*) also presents movement very close to the ground excitations, indicated by the 1.07 slope and the 87% correlation coefficient. The east mid-wall (*MW-T*) presents a small amplification of the ground displacements, indicated by slope of 1.42. Wall displacements as a function of air overpressure levels show more variability in the upper structure and mid-wall movements, as indicated by the lower correlation coefficients in the equations.

Table 1 shows computed differential wall displacements, shear, in-plane and mid-wall bending strains for the eleven events resulting in ground vibrations above the Brazilian regulations. Figure 5 presents correlations between in-plane tensile strain and bending strains with peak particle velocity. The radial motion of ground vibrations presets greater influence in the in-plane tensile strains in the east wall, parallel to the radial direction, than the transverse direction in the south wall. For bending strains, the racking motion of the transverse direction show greater influence in the east mid-wall bending strain than the direct hit of the radial ground velocities in the south wall bending strains.

The maximum computed differential wall displacement between upper and lower structure (*S2 – S1*) was 0.952 mm (0.037 in), computed for the blast at 33 m (108 ft) from the structure, resulting in a maximum shear strain (γ_{max}) of 377.7 micro-strains and in-plane tensile strain (ε_{Lmax}) of 139.7 micro-strains. Maximum bending strain (ε) was computed to be 123.1 micro-strains. These strain levels were compared to the failure strains for the cement grout used in between clay bricks, as mortar, and for wall coating. The cement grout was considered to be the weakest material in the wall construction and most likely to sustain cosmetic damages.

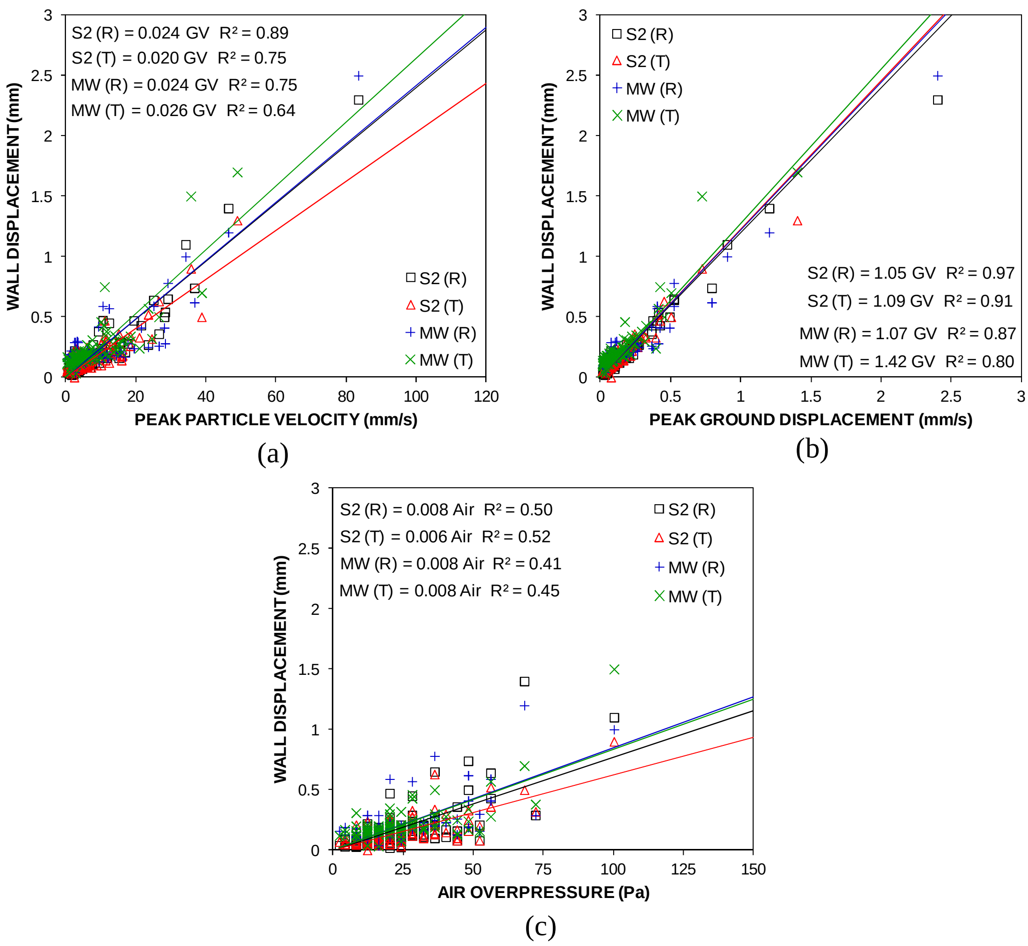
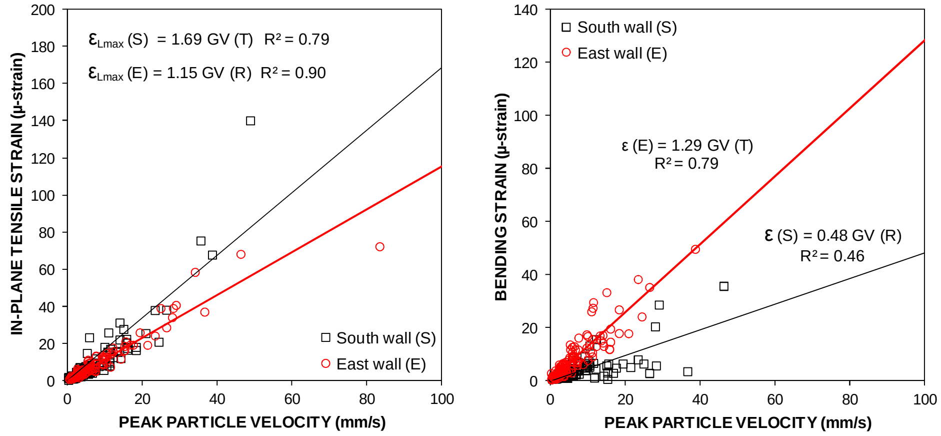


Figure 4: Horizontal displacement correlations between peak upper structure (*S2*) and mid-wall (*MW*) displacements with peak ground velocity (a), displacements (b) and air over-pressures (c) (note T and R represent transverse and radial components)

Table 1: Computed wall differential displacements, strain levels and safety factors against cracking for the highest blasts recorded at the structure.

Peak Particle Velocity (mm/s)		Differential Wall Displacement (mm)		Shear Strain (micro-strains)		In-plane Tensile Strain (micro-strains)		Mid-wall Bending Strain (micro-strains)		Safety Factor Against Cracking	
Radial	Transverse	South wall	East wall	South wall	East wall	South wall	East wall	South wall	East wall	In-plane Tensile	Bending
28.19	24.38	0.331	0.143	56.6	131.2	21.0	39.2	5.8	24.3	3.8	6.2
26.42	15.75	0.243	0.140	55.5	96.2	20.5	28.8	3.0	12.1	5.2	12.4
36.58	13.97	0.314	0.213	84.5	124.5	31.3	37.2	3.7	17.2	4.0	8.7
17.53	13.21	0.163	0.109	43.1	64.8	16.0	19.4	5.0	17.1	7.7	8.8
27.94	20.83	0.289	0.175	69.3	114.6	25.6	34.2	20.5	17.9	4.4	7.3
21.34	14.99	0.163	0.189	75.1	64.8	27.8	19.4	5.2	33.4	5.4	4.5
46.23	38.61	0.576	0.461	183.0	228.6	67.7	68.3	35.8	49.7	2.2	3.0
24.86	23.37	0.330	0.259	102.7	130.8	38.0	39.1	6.5	38.4	3.8	3.9
28.96	26.42	0.344	0.260	103.3	136.6	38.2	40.8	28.8	35.4	3.7	4.2
83.31	48.77	0.611	0.952	377.7	242.3	139.7	72.4	14.7	95.7	1.1	1.6
34.04	35.56	0.494	0.513	203.6	196.1	75.3	58.6	71.5	123.1	2.0	1.2



(a) (b)

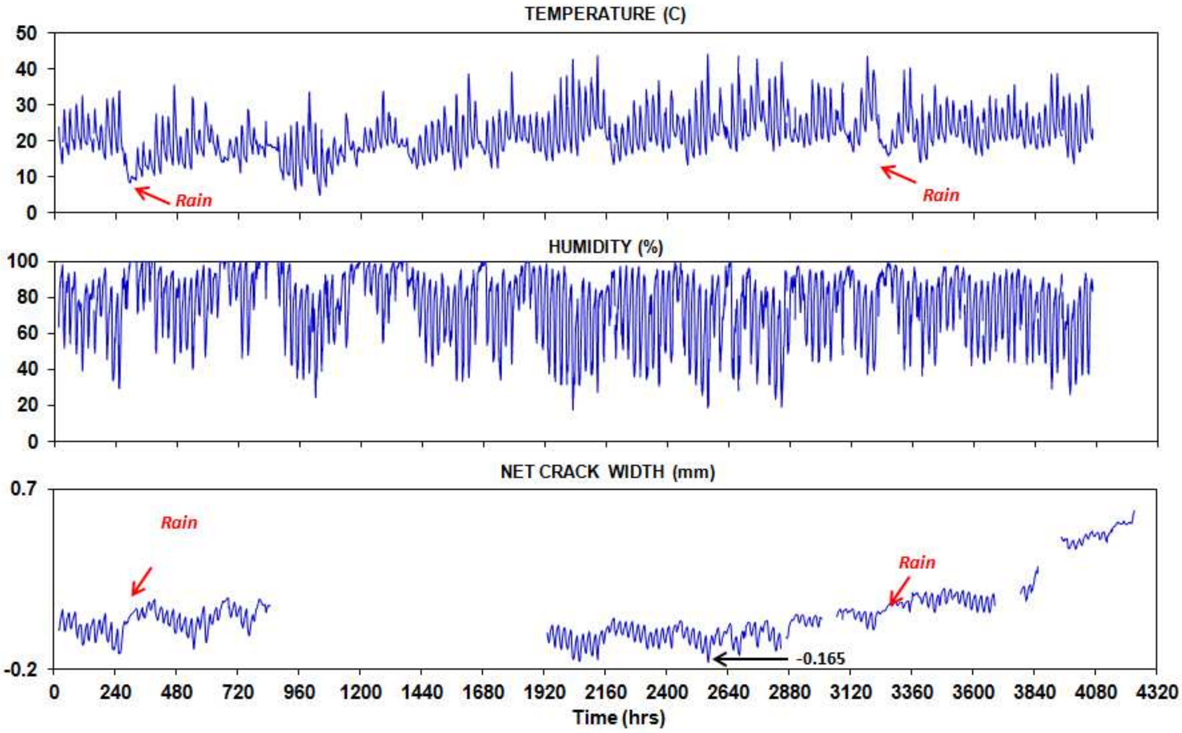
Figure 5: Peak particle velocity influence on induced in-plane tensile strain (a) and bending strains (b)

Tests conducted by Paes and Carasek (2002) and Silva and Capiteli (2006) in different compositions of cement grout presented failure by in-plane tensile strain raging from 150 to 250 micro-strains. Failure levels for bending strains were estimated to be around 250 micro-strains. The maximum computed in-plane tensile strain was 1.1 times smaller than the failure strain, while maximum computed bending strains for the test structure was 2 times the failure strain. Safety factors against cracking for the maximum strain levels resulting from the eleven blasts with highest PPVs are presented Table 1.

Figure 6 present long-term changes in crack width along with outside temperature and humidity variations for a period of 180 days (4320 hours). A separated plot for the end of the study is also shown in the figure. A portion of the crack data, as well as, temperature and humidity data were lost due to equipment malfunction, but this did not affect the quality of the remaining data.

Figure 6 shows that in general, long-term crack movement followed the trend in ambient humidity while short-term (or 24 hours) movement was consistent with diurnal temperature. When humidity increased, the crack opened (positive change) whereas a sudden increase in temperature resulted in crack closure. These effects are also evidenced by the scatter plots in Figure 7, where daily variations, or 12-hour cycles, in crack aperture (peak-to-peak) were correlated with daily variations in temperature (ΔT) and humidity (ΔH).

By the end of the study, mining was very close to the structure and the slope next to it had to be excavated in order to blast close-in to the house, shown in Figures 8 (a) and (b). Tension cracks were observed in the ground next to the house and extending underneath it. A new crack developed in the wall while the monitored crack and a pre-existing hairline crack, both widened, as seen in Figures 8 (c) and (d). The first blast close-in to the structure, at 32 m (105 ft), conducted after the slope was excavated, resulted in the growth of the new crack and closure of the monitored crack, shown by the sudden decrease in crack aperture in the end of study plot in Figure 6. When the slope was excavated and started to fail, the monitored crack widened (indicated by the dashed line in Figure 6) and after the blast, closed suddenly, remaining at this position during the rest of the study. Long-term crack movement analysis and dynamic crack response to blasting did not include this end-of-study response.



		Maximum Daily Variations		
	Overall Crack	Crack Width	Temp.	Hum.
Crack (mm)	0.841	**0.181**	0.133	0.148
Crack (μin)	33126	7107	5244	5831
Temp. (°C) Variation	39.2	15.1	**27.7**	26.1
Hum. (%) Variation	82.3	50.9	65.4	**69.2**

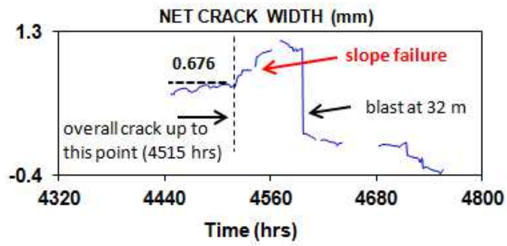


Figure 6: Weather-induced daily changes in crack aperture

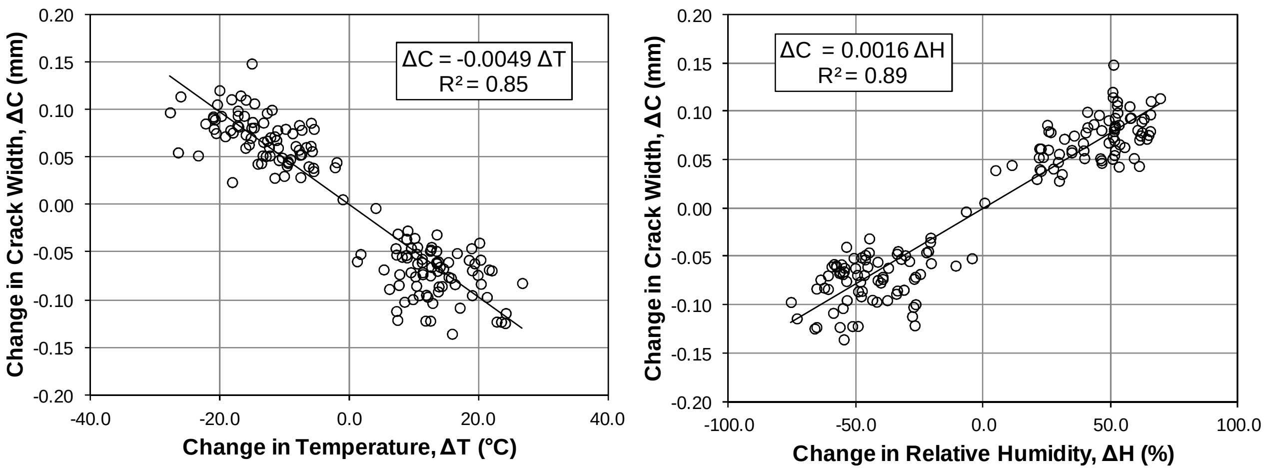


Figure 7: Daily variations in crack aperture correlated with variations in temperature and humidity

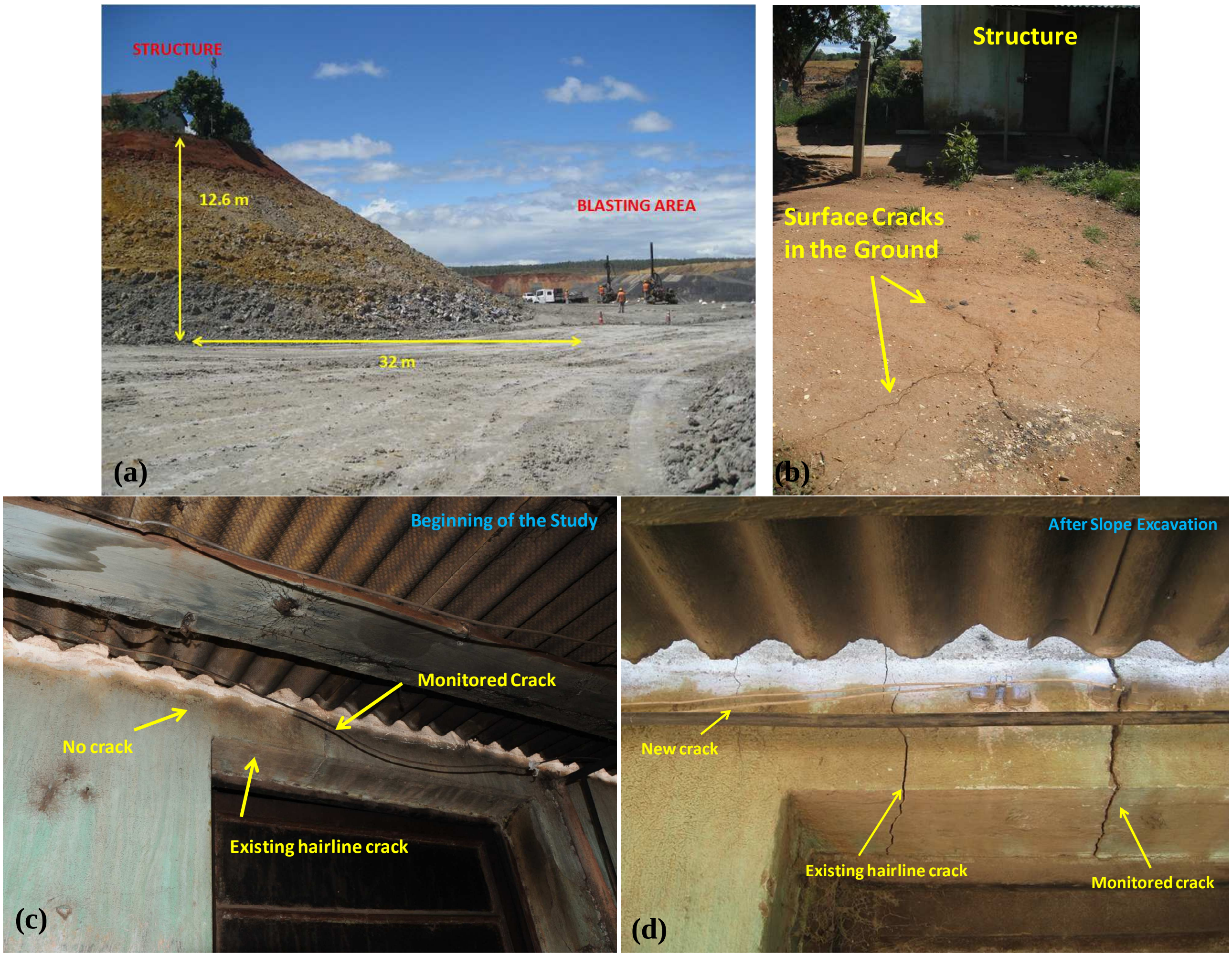


Figure 8: Slope excavated next to the test house (a), surface cracks in the ground near the house observed after slope excavation (b), photos comparing wall containing the monitored crack in the beginning of the study (c) and after the slope was excavated (d)

The total long-term opening of the crack was computed to be 0.841 mm (33,126 μin), while the maximum daily crack aperture was 0.181 mm (7,107 μin) at a maximum temperature variation of 15.1°C and maximum humidity change of 50.9%. The day of the maximum temperature variation, 27.7°C, the crack moved 0.133 mm (5,244 μin) and humidity varied 65.4%. Whereas, on the day of the maximum humidity variation, of 69.2%, crack gage aperture changed 0.148 mm (5,831 μin), and temperature varied 26.1°C. This indicates that the crack opening and closing has similar responses to daily variations of both temperature and humidity and may be more responsive to humidity changes. Weather fronts, such as rain, reduce the variation in crack movement, as indicated in Figure 6. The crack width remains nearly constant with little variation in temperature and humidity.

Blast-induced crack motions were recorded for 72 of the 115 events triggered by the ground sensor. Issues with the crack gage system prevented the recording of 43 blasts. Nonetheless, it was possible to evaluate crack response over a wide range of ground vibrations of 1.27 to 83.31 mm/s (0.05 to 3.28 in/s) with, peak frequencies between 3.9 and 36.5 Hz, and air overpressure levels from 2 to 484 Pa (0.0003 to 0.0702 psi)

Blast-induced crack aperture changes ranged from 0.089 mm (3,503 μin) to 0.309 mm (12,165 μin) peak-to-peak displacement. Figure 9 presents displacement time-histories comparing ground vibration levels (*GV*), upper structure motion (*S2*), crack displacement and air overpressure for the blast resulting in the largest crack width. The dashed line in the figure indicates crack responds closely with the longitudinal ground vibrations and upper structure movement in the south wall. The longitudinal direction is parallel to the east wall containing the crack and for this reason has a greater influence on the crack movement compared with the transverse direction, perpendicular to the east wall. Ground vibration levels shown in Figure 9 were 14.99 mm/s (0.59 in/s) with peak frequency of 23.2 Hz in the longitudinal direction and 13.97 mm/s (0.55 in/s) with peak frequency of 24.3 Hz in the transverse direction. Air over-pressure does not appear to affect crack movement.

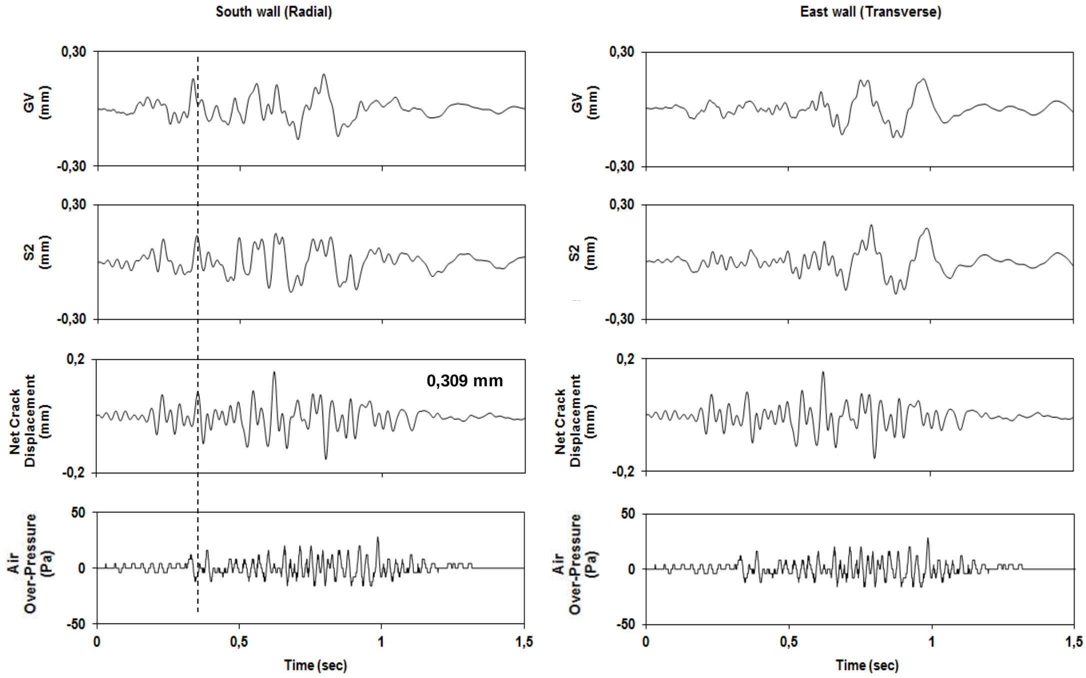


Figure 9: Waveforms comparing ground displacement (*GV*), upper structure movement (*S2*), crack displacement and air over-pressure pulse

Figure 10 shows correlations of changes in crack aperture with peak particle velocity, differential wall displacement and air overpressure showing evidence that both horizontal movements of the ground and the structure have similar influence on crack movement. This is indicated by the 67% correlation coefficient on Figure 10 (b). The 83% correlation coefficient between crack movement and the transverse ground velocity indicates that even though the transverse ground movement is perpendicular to the wall containing the crack, it has higher influence on crack width variation. Air over-pressure has very little influence in crack movement, shown by high scatter in the data.

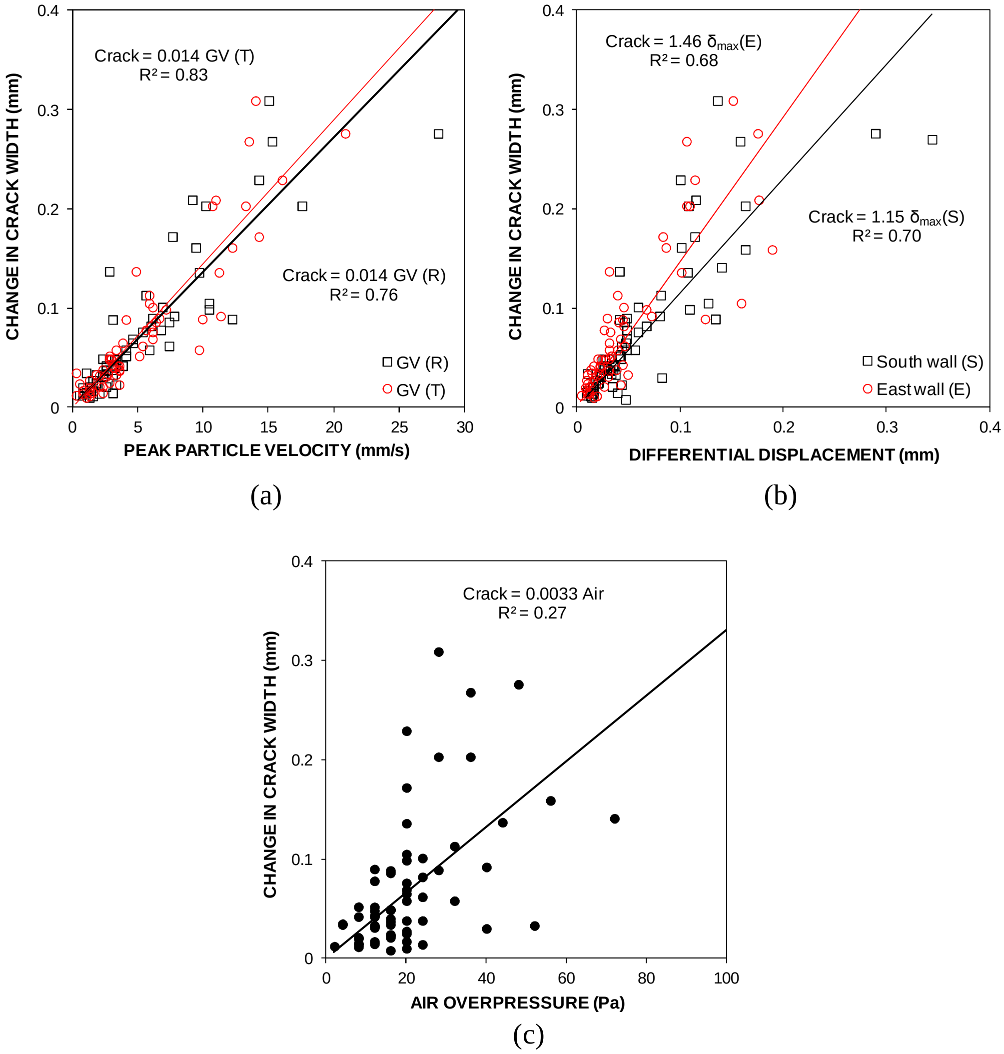


Figure 10: Crack response, peak-to-peak displacement, to peak particle velocity (a), differential corner movement (b) and air overpressure (c)

The maximum change in crack aperture of 0.309 mm (12,165 μin), resulting from blasting approximately 125 m (410 ft) from the structure before slope excavation and failure, was 2.7 times smaller than the long-term weather-induced overall crack displacement of 0.841 mm (33,126 μin). Figure 11 shows the maximum blast-induced crack time-history with the peak-to-peak displacement of 0.309 mm plotted within a 70 day period long-term weather-induced changes in crack aperture.

Seven blast events resulted in crack width changes greater than changes resulting from the maximum daily weather-induced variation. Peak ground vibrations of four of these events exceeded the Brazilian standard limits. All seven blasts were within 130 m (426.5 ft) from the structure with scaled distances ranging from 3.8 to 17.9 m/kg$^{1/2}$ (8.32 to 38.0 ft/lb$^{1/2}$). Even though these blasts resulted in crack movement greater than the maximum daily crack width change of 0.181 mm (7,107 μin), no extension or widening of the crack was observed. Blasting at the coal mine in the near future will be as close as

100 m (330 ft) from off-site structures in the vicinity of the mine. Blasts conducted over this distance resulted in maximum in-plane tensile strains of 26.1 micro-strains, while maximum bending strain was 26.9 micro-strains. These values are 5.7 times smaller than the limits for failure by in-plane tensile strains found in the literature and 9.3 times smaller than the failure limit by bending strains.

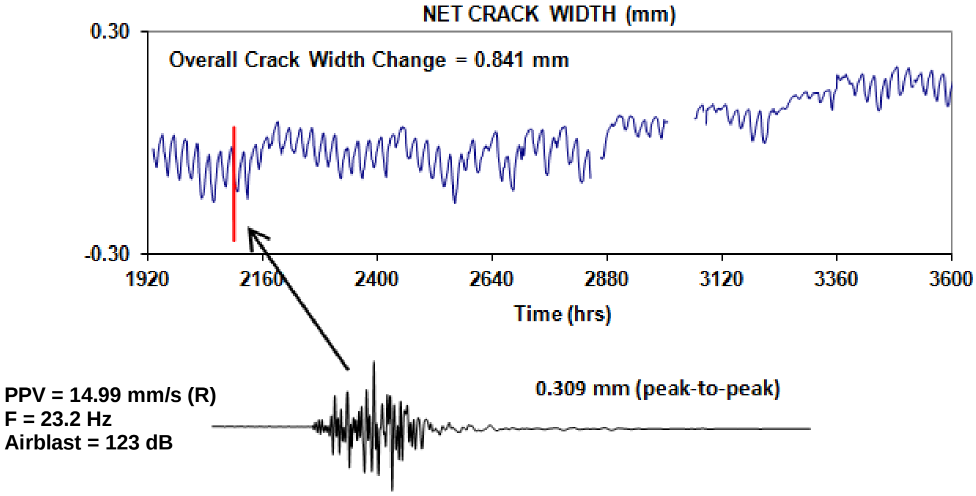


Figure 11: Comparison of daily variations in crack width from changes in temperature and humidity during a period of 70 days, with the maximum blast-induced crack movement

Conclusions

The following conclusions are drawn from this study:

- The mortar and brick structure was found to be rigid, with movement following very close the ground motions resultant from blasting activities
- Free response was not observed for this structure and the natural frequency of the structure could not be determined at this point.
- The strain levels computed for blasting events were lower than the failure limits for the weakest material comprising the structure. The stresses and deformations induced in the structure walls did not exceed the elastic limits of the materials and no permanent deformation was seen resultant from blasting activities. Therefore, cracks in the cement mortar wall coating, both in the exterior and interior of the structure, cannot be associated with blasting vibrations.
- A new crack was formed after the slope was excavated and was most likely caused by the static unloading of the foundation from the near-by slope failure. Therefore, the crack may have resulted from static foundation failure.
- Vibrations parallel to the wall containing the instrumented crack exerted greater influence in crack width changes while air overpressure did not presented significant influence on crack movement.
- The maximum blast-induced crack width change was 2.7 times smaller than the overall weather-induced long-term crack movement registered during the study. Seven blasts resulted in peak movement greater than the daily, 24 hr, influence by temperature and humidity. However, no new cracks were observed after these blasting events.

- The effects of climate can be considered as the biggest factor contributing to changes in crack aperture over time. The daily variation of temperature and humidity over long periods of time can fatigue materials comprising the structure resulting in superficial and cosmetic damages.
- Crack aperture variations are more sensitive to weather changes of temperature and humidity than to blast-induced motions. Residents living near blasting operations usually associate damages in the residences to blasting vibrations, however, in many cases ground vibrations and air overpressures from blasting are very low compared to the slow, long-term stimulus of the crack aperture from daily fluctuations in temperature and humidity. It is this daily cycle of opening and closing that produces high levels of stress at the end of the crack, provoking slow growth over time at the right conditions. It is not likely that blasting is the source of structure damage when compared with the influence of weather.

Acknowledgements

We would like to thank for the great support given by Copelmi Mineração Ltda, the coal mine company, for changing the mine design profile in order maintain the structure standing for a longer period of time. To Dr. Aimone-Martin for providing with the instrumentation systems and all support in the interpretation of the data. And finally to the Federal University of Rio Grande do Sul and to Dr. Enrique Munaretti for the seismographs and all support given during the study.

References

ABNT NBR 9653. 2005. Guide for the evaluation of effects of the use of explosives in mining and quarrying near urban areas – Proceedings. 9 p.

Aimone-Martin, C.T., Martell, M.A., L.M. McKenna, D.E. Siskind and C.H. Dowding. 2003. Comparative Study of Structure Response to Coal Mine Blasting, prepared for the Office of Surface Mining Reclamation and Enforcement, Pittsburgh. 209 p.

Aimone-Martin, C.T. and Eltschlager, K. E. 2003. Guidelines for Measuring and Evaluating Residential Structure Response, for the Office of Surface Mining and Reclamation Enforcement, Aimone-Martin Associates, LLC. Socorro, NM. 19 p.

Dowding, C.H. 2000. Construction Vibrations. Prentice Hall PTR

Dowding, C.H. 2008. Micrometer Crack Response to Vibration and Weather. International Society of Explosives Engineers. Cleveland, OH. 409 p.

Paes, I.N.L., Carasek, H. 2002. Desempenho das Argamassas de Rejuntamento no Sistema de Revestimento Cerâmico. Associação Nacional de Tecnologia do Ambiente Construído. v. 2, n. 2, p. 93-103.

Silva, N.G., Campiteli, V.C., 2006. Módulo de Elasticidade Dinâmico de Argamassa de Revestimento. Anais do 2º Encontro de Engenharia e Tecnologia dos Campos Gerais. Ponta Grossa, Paraná.

Siskind, D. E., Stagg, M.S., Kopp. J. W., and Dowding, C. H. 1980. Structure Response and Damage Produced by Ground Vibrations from Surface Blasting. Report of Investigation 8507, U.S. Bureau of Mines, Washington, DC. p. 79-82.